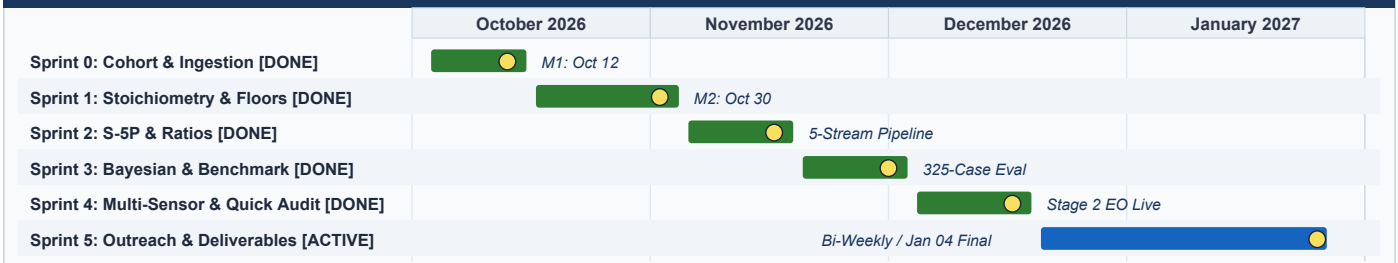
VeriCBAM: Capstone Project Work Plan, Risk Matrix & Execution Roadmap

A Data-Driven Decision-Support Tool for Screening Carbon Border (CBAM) Emissions Reports

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VeriCBAM Execution Gantt Chart & Milestone Timeline (Winter Semester 2026/2027)



1. Updated Execution Schedule & Milestone Status (Status: November 2026)

Sprint / Phase	Planned Dates	Key Completed Technical Tasks	Deliverable & Current Status
Sprint 0: Cohort & Ingestion	Oct 03 – Oct 14	Scoped project; ingested and cleaned 177 plant-years across 30 EU cement/steel facilities from EEA E-PRTR v16 and GEM trackers.	M1: Topic Approval (Oct 12) `[COMPLETED]`
Sprint 1: Stoichiometry & Planning	Oct 15 – Oct 26	Implemented calcination and steel reduction mass-balance floors (`cement_models.py`, `steel_models.py`); boundary gas-export rules.	M2: Proposal Approved (Oct 30) `[COMPLETED]`
Sprint 2: Satellite & Baselines	Oct 27 – Nov 04	Ingested Sentinel-5P TROPOMI NO□; built leave-one-year-out NO_x/CO□ ratios and one-sided robust pooled-MAD longitudinal drift scores.	5-Stream Data Pipeline `[COMPLETED]`
Sprint 3: Modeling & Evaluation	Oct 28 – Nov 05	Built Bayesian Log-Odds Fusion (Model E1) & GroupKFold Logistic Triage (Model E3); evaluated 325-case graded benchmark with 1,000 bootstrap CIs.	325-Case Benchmark `[COMPLETED]`
Sprint 4: Multi-Sensor & Dashboard	Nov 01 – Nov 05	Built live 8-tab Streamlit Cockpit with Tab 1 Quick Audit & PDF Dossier; integrated Stage 2 Multi-Sensor EO (Landsat 30m TIR, VIIRS 375m, Planet VHR).	Live App & Quick Audit `[COMPLETED]`
Sprint 5: Industry Outreach & Deliverables	Nov 06 – Jan 18	Real-world industry pilots (DNV, TÜV, DEHSt, importers), bi-weekly progress updates, presentation deck, walkthrough & pitch videos, and LinkedIn post.	Bi-Weekly Updates / Jan 04 Final `[ACTIVE]`

2. Project Risk Management & Resolution Matrix

Identified Technical Risk	Severity	Likelihood	Impact on Project	Implemented Mitigation & Current Resolution
Cloud Cover in Sentinel-5P Satellite	High	High	Trace-gas data gaps during northern European winter.	Resolved: Weighted satellite evidence by cloud fraction ($C_{sat} = 1 - f_{cloud}$) and integrated cloud-resilient thermal infrared (Landsat/VIIRS).
Off-Site Process Gases in Steelworks	High	Medium	False physical violations on blast furnace plants exporting gas to power plants.	Resolved: Mapped plant boundaries in GEM registry; set `stoich_applicable = False` for gas-exporting plants to eliminate false alarms.
Data Leakage in Machine Learning	Critical	Low	Artificially inflated accuracy from training and testing on the same plant.	Resolved: Enforced strict 5-Fold `GroupKFold(groups=facility_id)`, ensuring all years of any plant are strictly isolated between folds.
Confidentiality of Overseas Fraud	High	Medium	Real non-EU fraudulent CBAM reports are legally confidential.	Resolved: Built a 325-case controlled benchmark combining 177 real compliant years with 148 graded synthetic understatements (5%–50%).

3. Core Deliverables & Current Completion Status

1. VeriCBAM Python Engine (`src/`): 100% complete and passing 30/30 unit tests (`pytest`).
2. Interactive Cloud Decision Cockpit: Live at https://vericbam-uqpz4jrnpteqetfvjxtslh.streamlit.app/ featuring the new Tab 1 Quick Audit & 1-Click PDF Dossier Generator for zero-friction industrial demonstrations.
3. Stage 2 Multi-Sensor Satellite Dataset: Cached 30-facility dataset combining USGS Landsat-8/9 30m TIR Δ T, NASA FIRMS VIIRS 375m daily operating calendars, and Planet VHR freight contrast.
4. Industry & Authority Outreach Suite: Standalone 1-page Quick User Guide (`06_Quick_User_Guide.pdf`) and standardized demonstration protocol for verification bodies (DNV, TÜV, Bureau Veritas) and customs authorities (DEHSt).

4. Bi-Weekly Progress Roadmap & Final Deliverables Schedule (Winter 2026/2027)

- Bi-Weekly Progress Cadence: Progress updates submitted to Dr. Humera Noor every two weeks: Nov 09 (Update 1: Live Tab 1 Quick Audit & Stage 2 Multi-Sensor benchmark), Nov 23 (Update 2: Industry pilot feedback & automated PDF dossier validation), Dec 07 (Update 3: Multi-sector sensitivity & deck structure), and Dec 21 (Update 4: Video walkthrough production & LinkedIn draft).
- Final Deliverables Package (Jan 04, 2027):

1. Presentation Deck: Executive slide deck presenting operational ROI and physics-grounded AI methodology.
2. Long-Form Video (15–20 min): Technical walkthrough of the multimodal data pipeline and live interactive audit.
3. Elevator Pitch Video (1–2 min): Concise executive video demonstrating rapid declaration screening for CBAM importers.
4. LinkedIn Post: Public industry release driving adoption among EU climate verifiers and customs authorities.
5. Open-Source Codebase & Live App: Reproducible GitHub repository and live Streamlit cloud decision cockpit.